

Stratified And Hybrid Risk-Limiting Audit Structures

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Abstract

Stratified and hybrid risk-limiting audits are needed when one contest spans more than one evidence surface. A contest may have ballot-level CVRs in one stratum, only ballot-polling evidence in another, or batch-level comparison evidence in a third. RCOUNT now records that SUITE-style coordination layer as a replayable structure: a stratified coordinator names its combining rule, nuisance parameter, component runs, stratum methods, ballot counts, source references, and risk allocations. The implemented verifier checks that the strata are not flattened, that allocations form a full partition of the risk budget, and that every stratum points to an existing non-stratified component run. It deliberately reports the combined statistical risk replay as a boundary, not as implemented math.

1 Introduction

Some election audits are not homogeneous. One part of a contest may have ballot-level CVRs and support ballot comparison. Another part may only support ballot polling. A third part may be naturally audited as a batch comparison. The public risk statement still concerns one contest outcome, so the audit record needs a coordinator that preserves the stratum boundaries and combines the independent stratum evidence under a declared rule.

RCOUNT now records that coordinator for the SUITE and hybrid RLA layer. The implemented object is not a combined-risk calculator. It is a structure contract: the package must show which component audit run belongs to each stratum, which method that stratum uses, what allocation was assigned, and which combining and nuisance-parameter fields define the intended statistical surface.

This paper covers the implemented V.17 slice. It claims that a record with method id `stratified-hybrid-rla-v1` is well formed, replayable as a declared structure, and protected against flattening into a false single-frame audit. It does not claim that RCOUNT has implemented the final SUITE P-value combination or any other combined statistical risk calculation.

2 Stratified Contract

The coordinator is an `AuditAlgorithmRun` whose method id is `stratified-hybrid-rla-v1`. It is a boundary-only record: the component runs carry their own audit method evidence,

while the coordinator records how those components are meant to be interpreted as one stratified audit.

Field	Meaning
<code>method_id</code>	Versioned coordinator name. Current value: <code>stratified-hybrid-rla-v1</code> .
<code>combining_rule_id</code>	Declared rule for combining independent stratum evidence, such as the synthetic <code>suite-nuisance-boundary-v1</code> fixture rule.
<code>nuisance_parameter</code>	Rational nuisance value or optimization transcript anchor required by the declared design.
<code>risk_limit_ppm</code>	Contest-level risk limit, stored in parts per million when present.
<code>strata</code>	Ordered stratum records. The implemented verifier requires at least two strata.
<code>decision</code>	Boundary decision for the coordinator until combined-risk replay is implemented.

Each stratum record carries the public link from the coordinator to one component audit run:

Field	Meaning
<code>stratum_id</code>	Stable, nonempty stratum name. Stratum ids must be unique within the coordinator.
<code>method_id</code>	Non-stratified audit method used by the component run.
<code>component_run_id</code>	Existing audit algorithm run that supplies the stratum-level evidence.
<code>ballot_count</code>	Optional stratum sampling-frame size; if present it must be positive.
<code>allocation_ppm</code>	Required positive allocation share. The coordinator allocations must sum to 1000000.
<code>source_refs</code>	Optional evidence references for the stratum.

The partition rule implemented today is therefore structural. The strata must remain distinct, must refer to existing component runs, must not recursively name another stratified coordinator as their method, and must allocate the entire one-million-parts risk budget. The statistical combination is preserved as declared metadata rather than recomputed.

3 Implemented Surface

The core verifier calls `verify_stratified_hybrid_design` for every audit algorithm run. For non-stratified methods, the function rejects stray stratified fields: `combining_rule_id`, `nuisance_parameter`, or a nonempty `strata` list. For the stratified method id, it enforces the coordinator shape before the general sample-step checks continue.

Check	Failure caught
<code>STRATIFIED_HYBRID_RLA_METHOD_ID</code>	Unsupported or misspelled coordinator method ids are rejected by the audit-algorithm method registry.
<code>combining_rule_id</code>	Missing or blank combining rule on a stratified coordinator.
<code>nuisance_parameter</code>	Missing nuisance parameter or rational value without a positive denominator.
<code>strata.len()</code>	Flattened or empty coordinator with fewer than two strata.
<code>assertions, sample_steps</code>	Coordinator incorrectly carrying direct assertion/sample-step evidence instead of component references.
<code>stratum_id</code>	Empty or duplicate stratum ids.
<code>component_run_id</code>	Self-reference or reference to a missing component run.
<code>method_id</code>	Unsupported stratum methods and recursive stratified stratum methods.
<code>ballot_count</code>	Zero ballot count when a count is declared.
<code>allocation_ppm</code>	Missing, zero, oversized, or non-partitioning allocations; the sum must be 1000000.

The audit replay crate exposes the same method id through `replay_audit_algorithm_statistics`. Its current behavior is an explicit boundary transcript: `stratified/hybrid combined-risk replay` is recorded but not replayed. That surface is important because it lets a public artifact name the intended combined-risk method without implying that RCOUNT has recomputed the SUITE or hybrid risk calculation.

4 Fixtures

The positive fixture is `synthetic_stratified_hybrid_package`. It starts from the synthetic batch-comparison algorithm package, adds the synthetic Minerva multi-round ballot-polling run, and then adds a coordinator run named `audit-run:stratified-hybrid-boundary`. The coordinator uses `stratified-hybrid-rla-v1`, declares the combining rule `suite-nuisance-boundary-v1`, records nuisance parameter 1/2, and sets the decision to `Boundary`.

Fixture	Role
<code>synthetic_stratified_hybrid_package</code>	Positive two-stratum coordinator: one batch-comparison stratum with 80 ballots and one Minerva ballot-polling stratum with 60 ballots. Each stratum receives a 500000 ppm allocation and points to an existing component run.
<code>synthetic_bad_stratified_hybrid_package</code>	Missing-component negative fixture. It edits the ballot-polling stratum to reference <code>audit-run:missing-stratum</code> , causing <code>MissingStratifiedHybridComponent</code> .
<code>synthetic_flattened_stratified_hybrid_package</code>	Flattening negative fixture. It truncates the coordinator to one stratum and gives that stratum the full 1000000 ppm allocation, causing <code>InvalidStratifiedHybridDesign</code> .

The core tests check the same contract directly. The positive test confirms that the package verifies, that the stratified run has two strata, that the combining rule and nuisance parameter are present, and that both allocations are 500000. The negative tests confirm that a missing component and a flattened one-stratum coordinator are rejected.

The audit crate adds a replay-surface test for the positive package. It finds the `stratified-hybrid-rla-v1` run, calls `replay_audit_algorithm_statistics`, and confirms that the transcript status is `Boundary` with a message containing `combined-risk replay`.

5 Boundaries

The V.17 claim boundary is narrow. RCOUNT preserves the stratum structure, component run links, combining-rule id, nuisance parameter, and allocation partition. It rejects malformed structures that would make a hybrid audit look like one homogeneous sampler or one unsupported statistical run.

It does not yet replay the combined statistical risk. The plan states this directly: the combined risk claim depends on correct stratum definitions and sampling frames, and RCOUNT now preserves strata and component run references, but still does not claim combined-risk replay. The audit crate mirrors that boundary by returning a method-specific boundary transcript instead of a pass or fail statistical transcript for the combined risk.

This also means that SUITE, SHANGRLA/ALPHA, Fisher combination, and any state-specific hybrid formula remain outside the implemented result unless they are present in the component methods that already replay. V.17 is the coordinator substrate on which those calculations can be added without losing the public record of which ballots, batches, or jurisdictions belonged to which evidence surface.

6 Conclusion

RCOUNT now has a replayable structure contract for stratified and hybrid RLAs. The verifier recognizes `stratified-hybrid-rla-v1`, requires a declared combining rule and nuisance parameter, verifies that at least two strata remain distinct, checks that each stratum points to an existing non-stratified component run, and requires allocation shares to form a complete 1000000 ppm partition.

The implementation is intentionally understated. It protects the public audit record from missing components and flattened-stratum misuse, and it exposes the method through algorithm replay as a documented combined-risk boundary. The next statistical layer can implement the selected SUITE or hybrid combination rule on top of that structure; until then, RCOUNT claims structure verification, not combined-risk replay.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Replayable risk-limiting audit samplers and stopping checks. Technical report, Apportionment Research, 2026b.